2- Mirrors[29,30] and lenses[30]:

2-1 -Introduction:

*Mirrors and lenses are every where-in home, auto headlamps, telescope, laser, human eye and many other places.

*Plane and spherical mirrors are used to form three dimensional images of three-dimensional objects.

*If the size, orientation and location of an object relative to a mirror or a lens are known, the law of reflection and ray tracing can be used to locate the image graphically.

2-2-Images formed with plane mirrors: Fig.(2-1 and 2-2 –a,b,c and d)

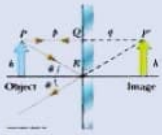


Fig.(2-1)

Images with mirrors are formed under some conditions:

1-Nonparallel rays from any source are reflected from the mirror surface ,converge, and form a corresponding image object(point, extended object, different size and different volume,….).

2-The law of reflection ensure that for example from Fig.(2-2 -a): the triangle SNP and S/NP are equal, S S/ is normal to the mirror, SN= S/N and The image is said to be a virtual image.

The properties of the constructed images :

1-The image is constructed at a distance behind the mirror = as the object is in front.

2- The image is unmagnified, virtual and upright.

3- The image having an apparent left right reversal.

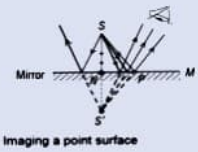
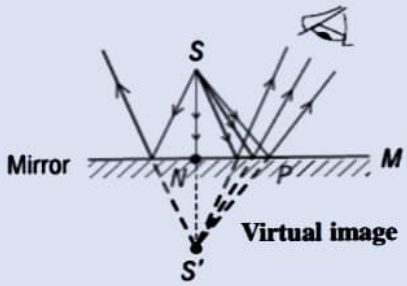
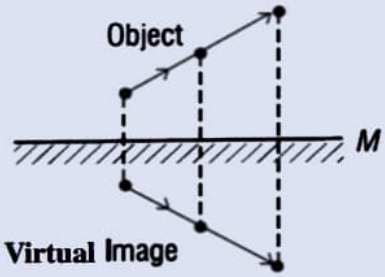


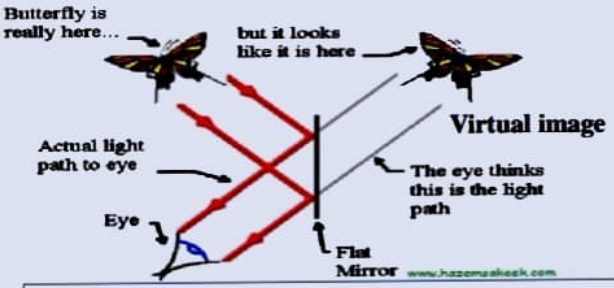
Fig.(2-2- a)



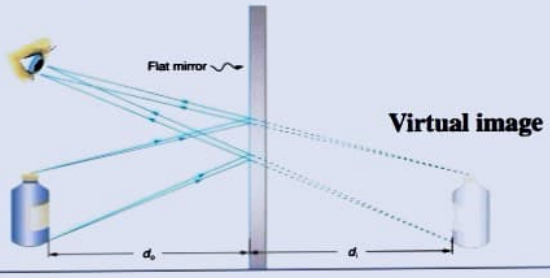
(a) Imaging a point surface



(b) Imaging an extended object



(c) Image is the same size.



(d) Image is the same volume.

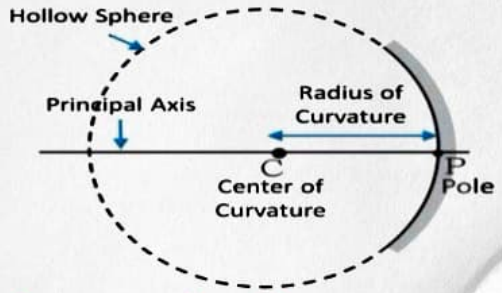
Fig.(2-2- a, b, c and d)

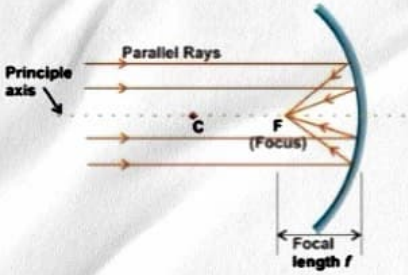
2.3- Images formed with spherical mirrors

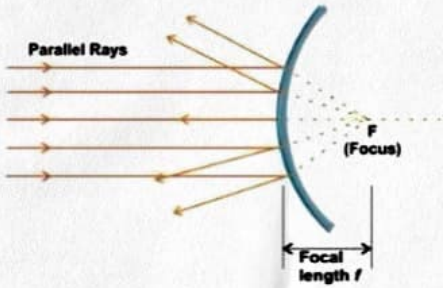
Mirrors as a part of

a) **Concave Mirror**

b

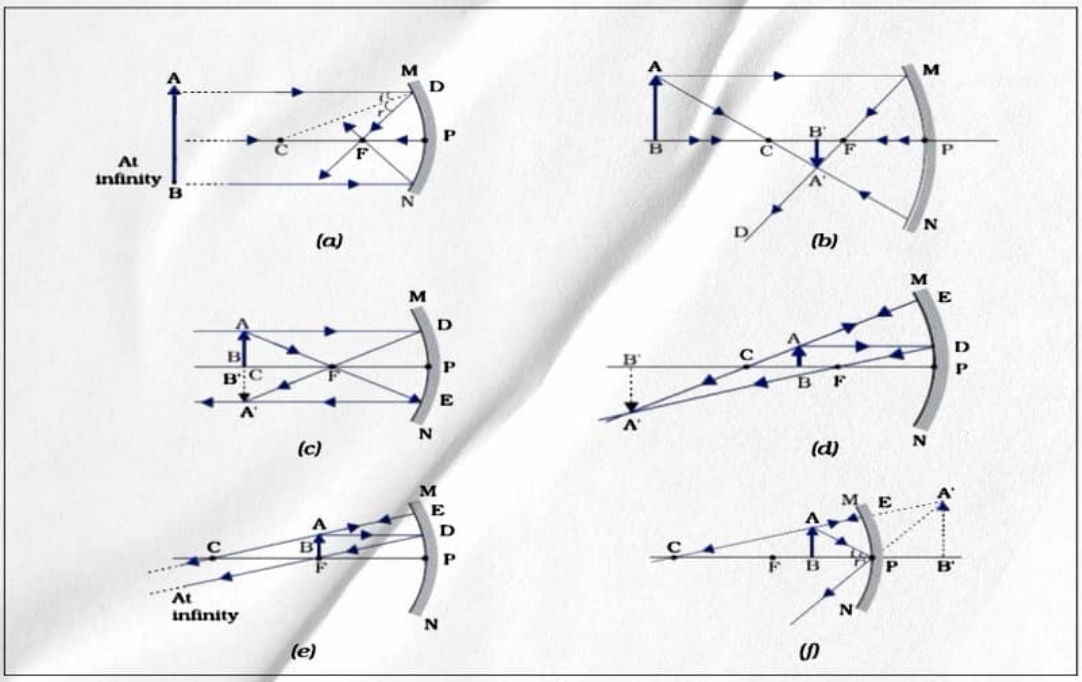






c-The images in the concave mirror ray diagram.

d-The images in t



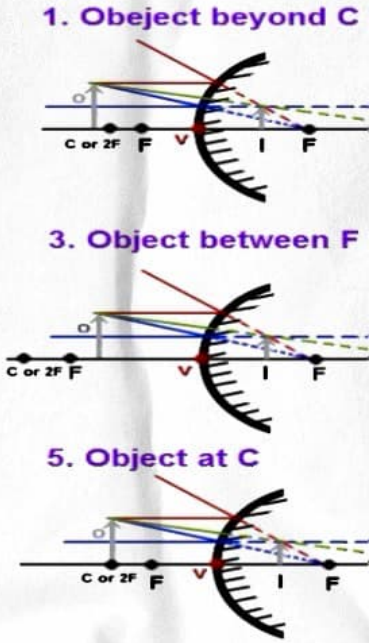
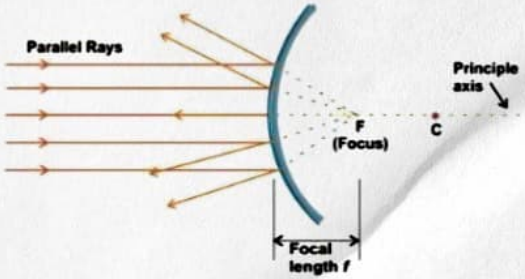


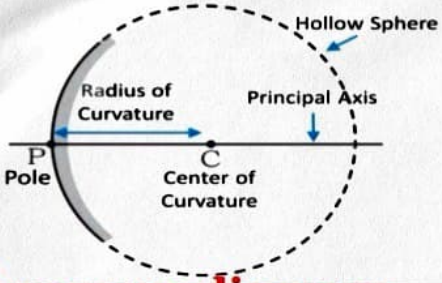
Fig.(2-3- a , b , c and d)

with spherical mirrors **Fig.(2-3 a , b , c and d):**

s a part of Sphere

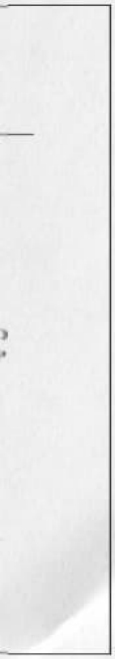
b) Convex Mirror





agram.

d-The images in the convex mirror ray diagram.



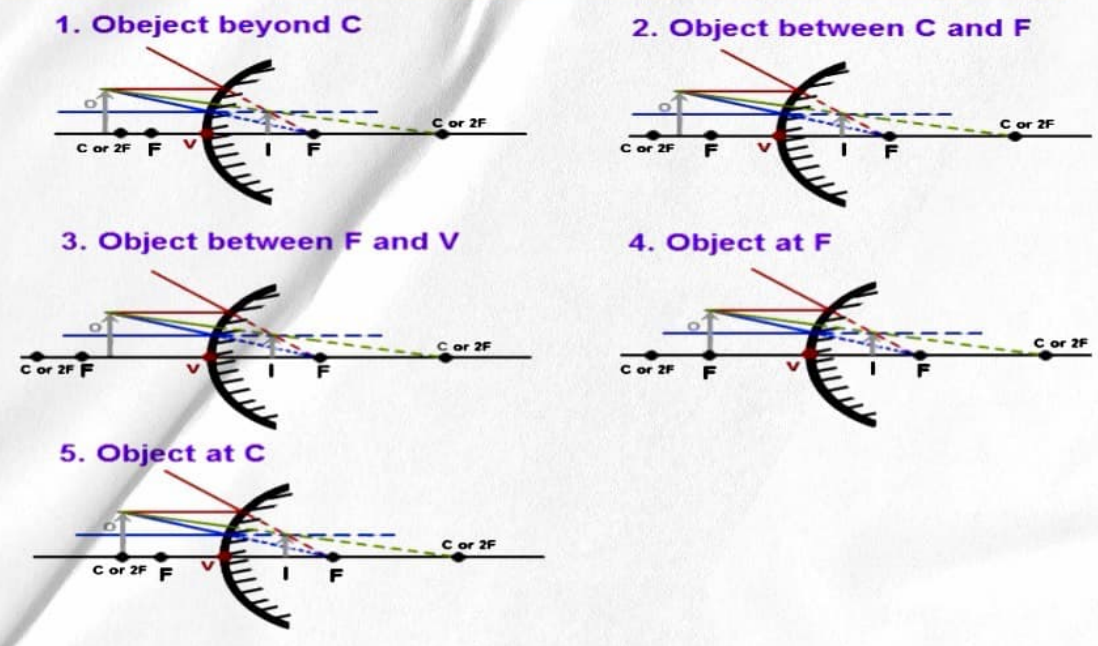
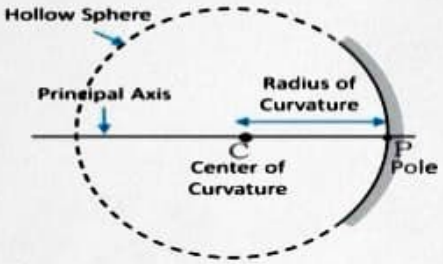


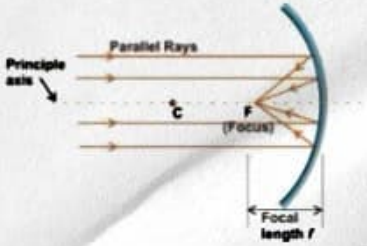
Fig.(2-3- a , b , c and d)

2.3- Images formed with spherical mirrors Fig.(2-3 a , b , c and d):

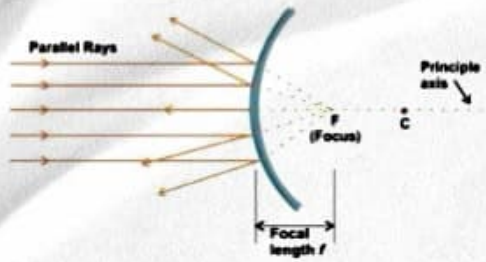
Mirrors as a part of Sphere

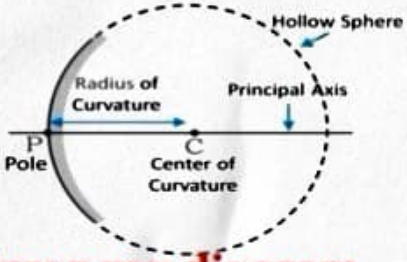
a) Concave Mirror



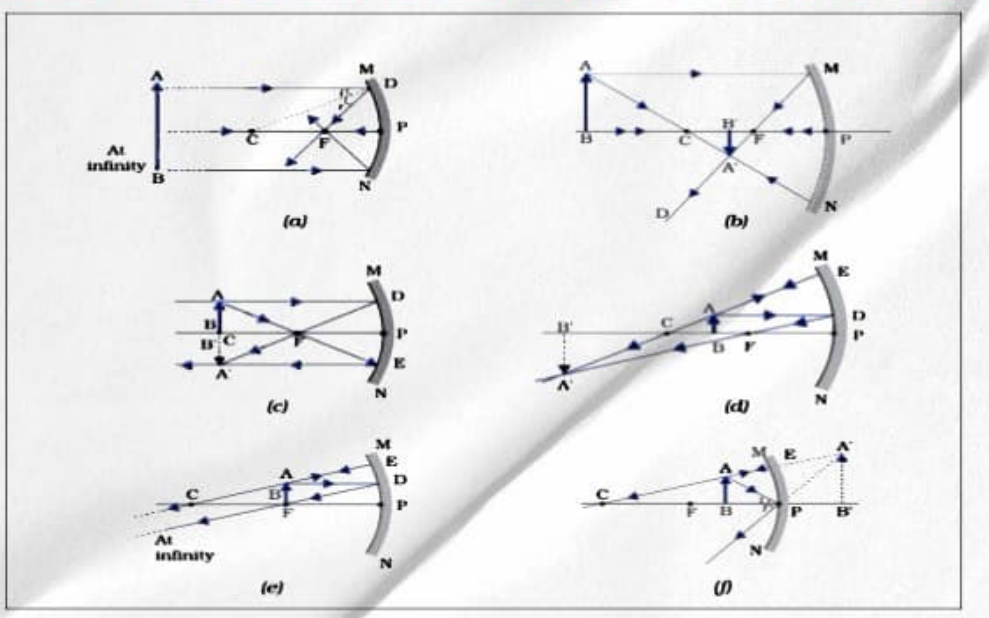


b) Convex Mirror





c-The images in the concave mirror ray diagram.



d-The images in the convex mirror ray diagram.

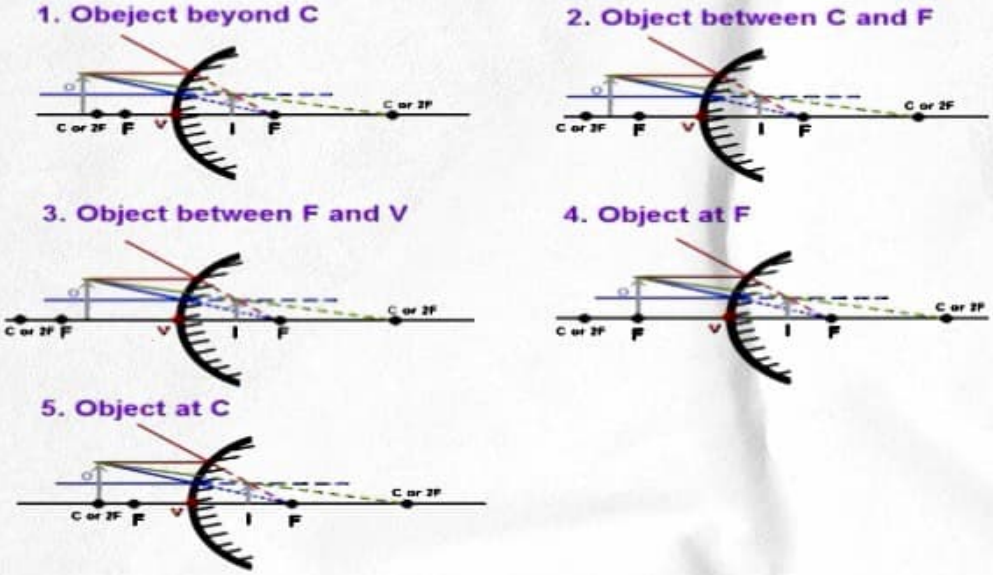
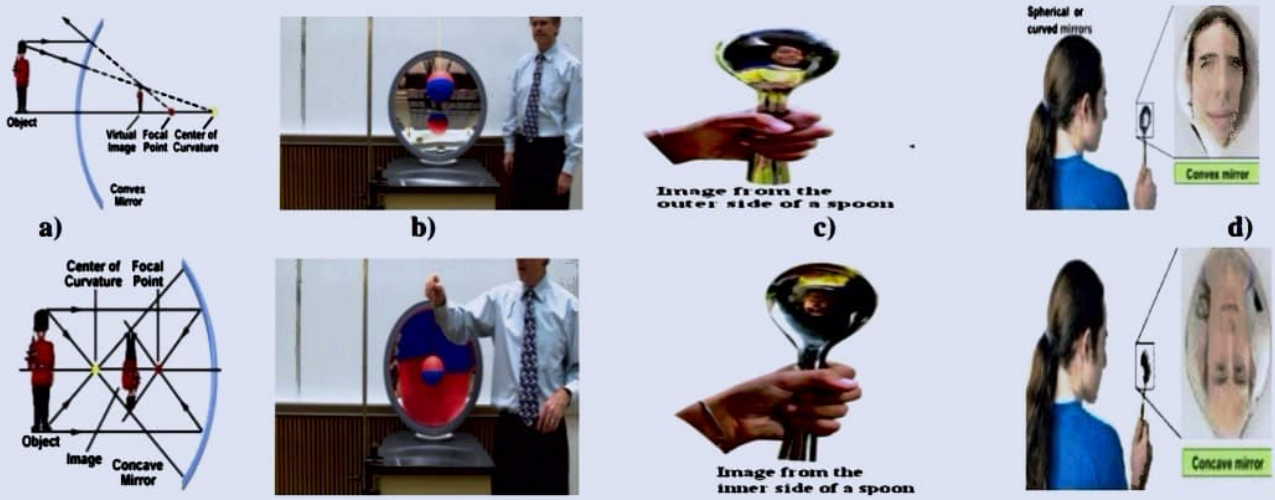


Fig.(2-3- a , b , c and d)

2-3-1-Comparison between the image properties of the convex[31] and concave[36] mirror. (Fig.2-4-a,b,c and d)



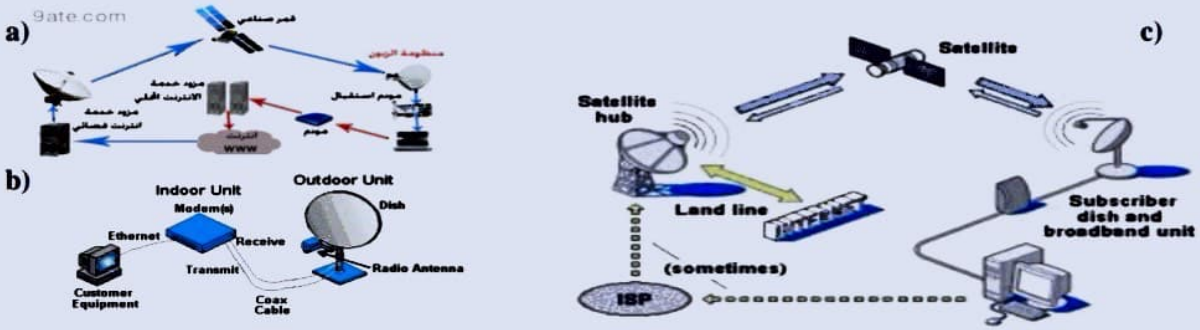
(Fig.2-4-a,b ,c and d)

2-3-2-General applications.(Fig.2-5-a,b and c)

Important notes:

1-The mirrors laws are used by the scientists to invent a numerous applications , such as the pictures from the Satellites and Radar and……. .

2-The dishes in theses electronic circuits are playing as a sender or a receiver for the electromagnetic waves.



a) and b)- Internet from the Satellite. c) Connection systems trough the Satellite.

(Fig.2-5-a,b and c)

2-4-The General equation of the mirrors[32] .Fig.(2-6)

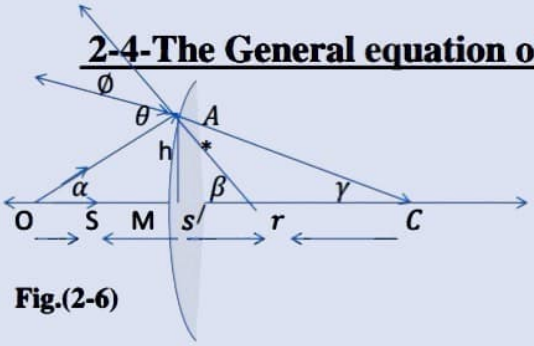


Fig.(2-6)

From figure , Fig.(2-6)

the angles is very small then tan $\theta = \theta$

The incident angle:

$\theta = \alpha + \gamma$ (Δ OAC)

The reflection angle:

$\emptyset = *$ (C A r)

$\beta = * + \gamma$ *in* (Δ C A r)

$\emptyset = \beta - \gamma$

But The law of reflection:

$\theta = \emptyset$

From the Figure Fig.(2-6) :

$\alpha + \gamma = \beta - \gamma$ $\quad 2\gamma = \beta - \alpha$

$$2\frac{h}{r} = \frac{h}{s/} - \frac{h}{-s}$$

$$\frac{1}{s/} + \frac{1}{s} = \frac{2}{r}$$

Special case:

1- When the object is put in ∞ the image will be in the focus.

2- The distance between the focus and the pole of the mirror is the focal length.

$$\frac{1}{s/} + \frac{1}{s} = \frac{2}{r}$$

$$\frac{1}{\infty} + \frac{1}{f} = \frac{2}{r}$$

$f = \frac{2}{r}$, ...r =2f

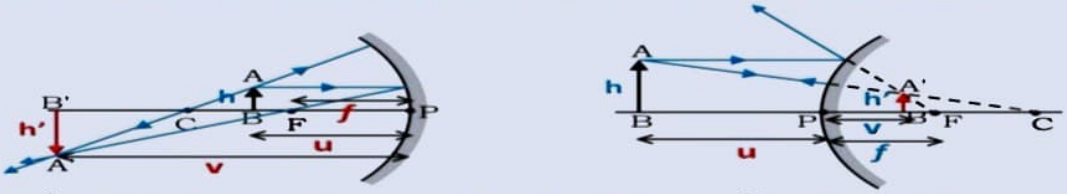
$$\frac{1}{s/} + \frac{1}{s} = \frac{1}{f}$$

The equation of the concave mirror:

$$\frac{1}{s/} + \frac{1}{s} = \frac{1}{f} = \frac{2}{r}$$ and $$\frac{1}{s/} + \frac{1}{s} = \frac{1}{f} = -\frac{2}{r}$$

The equation of the convex mirror

2-4-1- Mirror Formula and Magnification



a) Concave Mirror (Fig.2-7-a and b) b) Convex Mirror

Mirror Formula: $$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Magnification $$m = \frac{Height\ of\ image}{Height\ of\ Object}$$

$$m = \frac{-v}{u}$$

2-4-2- Sign Convections for Mirrors. Table 2-1

Quantity	Symbol	In Front	In Back	Upright Image	Inverted Image
Object location	s	+	−		
Image location	s'	+	−		
Focal Length	f	+	−		
Image height	h'			+	−
Magnification	M			+	−

Table 2-1

Sign convention:

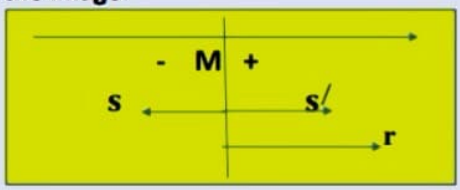
1-S and s' → **+ when it is located at the left of the vertex.** → **Real object and mirror**
→ **- when it is located at the right of the vertex.** → **Virtual object and mirror**

2-radius r → **+ for concave mirror.**
→ **- for convex mirror .**

3-Vertical dimension : → **+ Above the optical axis .**
→ **- Blow the optical axis.**

Examples:

Example (2-1): An object is placed in front of the mirror at a distance (75 cm) on the principal axis of a convex mirror , its radius of curvature equal to(100 cm).Find the place of the image.



Solution 2-1):

$\frac{1}{s'} + \frac{1}{s} = \frac{1}{f} = \frac{2}{r}$ .....r = 100 cm , S=-75 cm ,

$\frac{1}{s'} + \frac{1}{s} = \frac{2}{r}$ ..... $\frac{1}{s'} = \frac{1}{50} + \frac{1}{75} = \frac{125}{3750}$ ..... $s' = \frac{3750}{125}$ =30cm

Example (2-2):

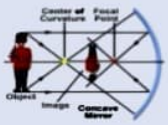
An object is placed in front of the mirror at a distance (15 cm) on the principal axis of a concave mirror , its radius of curvature equal to(40 cm).Find the place of the image.

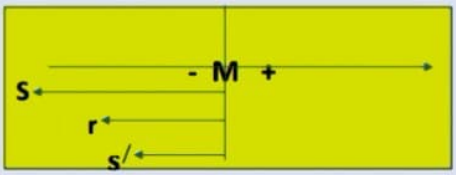
Solution 2-2):

$\frac{1}{s'} + \frac{1}{s} = \frac{1}{f}$

f= -20 cm , s= -15 cm

$\frac{1}{s'} - \frac{1}{15} = -\frac{1}{20}$ ..... $\frac{1}{s'} = -\frac{1}{20} + \frac{1}{15} = \frac{-15+20}{300} = \frac{1}{60}$





Example (2-3):

An object is placed in front of the mirror on the principal axis of a concave mirror, its focal length equal to(10 cm).If the place of its real image is at a distance (15cm) .Find the place of the object.

Solution(2-3):

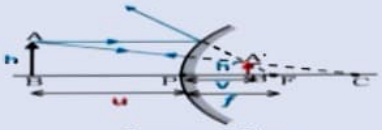
$\frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f}$ ..... $s^{/} = -15$ cm , $f = -10$ cm

$\frac{1}{-15} + \frac{1}{s} = \frac{1}{-10}$ ..... $\frac{1}{s} = \frac{1}{-10} + \frac{1}{15} = \frac{-5}{150}$

S=-30 cm

Example: (2-4)

- S - ⟶
- $s^{/}$ - ⟶
- r - ⟶



Convex Mirror

An object of length (1.5 cm) is placed on the principal axis of a convex mirror , its radius of curvature equal to(30 cm). Find the place
and the length of the image , if the object is at the right of the vertex (the pole of that mirror ..... i.e., ... S -) by a distance (25 cm) ?

$$\frac{2}{r} = \frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f}$$

Solution (2-4):

Apply the equation and use the sign rules.

$\frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f} = \frac{2}{r}$ ∴ s =- 25 cm , r = - 30 cm

$\frac{1}{s^{/}} + \frac{1}{-25} = \frac{2}{-30}$ ∴ $\frac{1}{s^{/}} = + \frac{1}{25} - \frac{2}{30} = \frac{(2x25)+(-30)}{(-30)(25)}$ ..... $s^{/} = \frac{(25)(-30)}{(50)+(30)} = -37.5$ cm

This result means that the image is located at left to pole of the mirror , this image is real, and it can be formed on a screen.

$m = \frac{-s^{/}}{s} = \frac{h^{/}}{h}$, $\frac{37.5}{-25} = -1.5$ ..... $h^{/} = -(1.5)(1.5) = -2.25$ cm

This result means that ,the length of the image is (2.25 cm), and it is placed blow the optical axis (this means that the image is inverted).

Example:(2-5)

An object 3cm high is placed 20 cm from a convex mirror with a focal length of magnitude 8cm. find (a) the position of the image.(b) The magnification of the mirror, and (c) the height of the image.

Solution (2-5): Apply the mirror equation and use the sign rules. (s, $s^{/}$, r ,f are -)

$\frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f} = \frac{2}{r}$

(a) $\frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f}$ $\frac{1}{s^{/}} + \frac{1}{20} = \frac{1}{-8}$ $s^{/} = -5.71$ cm

(b) $m = \frac{s^{/}}{s} = \frac{h^{/}}{h}$ $m = \frac{-s^{/}}{-s}$ $m = \frac{-5.71}{-20} = 0.286$

(c) $m = \frac{h^{/}}{h}$ $h^{/} = m\,h$ $h^{/} = (3)(0.286) = 0.858$ cm

Example:(2-6)

When a women stands with her face 40 cm from a cosmetic mirror, the upright image is twice as tail as her face. What is the focal length of the mirror? (s, $s^{/}$, r ,f are -)

Solution (2-6):

$m = \frac{h^{/}}{h}$ $m = -\frac{h^{/}}{h} = -2$ (- means inverted image)

$h^{/} = -2\,h = -2\,(40) = -80$ cm.

$\frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f}$ $\frac{1}{40} - \frac{1}{80} = \frac{1}{f}$ (- when it is located at the right of the vertex).

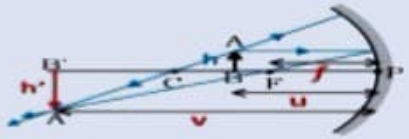
$f = 80$ cm.

Example (2-7)

$S \longrightarrow -$

$s^{/} \longrightarrow -$

$r \longrightarrow +$



Concave Mirror

An object of length (1.5 cm) is placed on the principal axis of a concave mirror , its radius of curvature equal to(30 cm). Find the place and the length of the image , if the object is at the right of the vertex (the pole of that mirror ..i.e., ... S (-) by a distance (25 cm) ?

Solution (2-7): Apply the equation and use the sign rules.

$\frac{1}{s^{/}} + \frac{1}{s} = \frac{1}{f} = \frac{2}{r}$, s= -25 cm , r= 30 cm ..... $\frac{1}{s^{/}} - \frac{1}{25} = \frac{2}{30}$..... $s^{/} = 9.4$ cm

This result means that the image is located at left to pole of the mirror ,this image is virtual, and it can not be formed on a screen.

$m = \frac{-s^{/}}{-s} = \frac{-9.4}{-25} = 0.38$ ..... $h^{/} = (1.5) \times (0.38) = 0.57$ cm

This result means that the value is positive and smaller than one ,I .e. the image is virtual, smaller and upright.

2-4-3-Uses of convex mirror[31]

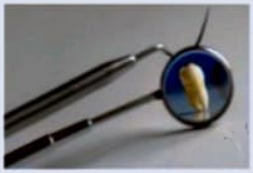
1-The convex mirror is used as side-view mirror on the passenger's side of a car because it forms an erect and smaller image for the way behind the car .

2-The convex mirror is suitable for convenient shop and big supermarket and any other corner where need anti-thief , It is used in the turning off the road and parking .

3-Convex mirrors are used inside the buildings , They are also used in making lenses of sunglasses , They are used in magnifying glass , They are used in securities, and they are used in telescopes .

4-They can be used as streetlight reflectors because they can spread the light over a bigger area , They are put on the corners of roads so that you can see any cars coming to avoid the collisions and they are used as ceiling dome mirrors .

5-The convex mirror Is used by the dentist to see the lower parts of the dental.



6-It Is used in receiving radar and T.V. waves.

2-4-4-Special application on convex mirror .(Fig. 2-8-a,b,c,d,e and f.)

a)

b)

c)

d)

e)

f)

(Fig. 2-8-a,b,c,d,e and f.)

2-4-5-Uses of the concave mirror[36]:

1-The concave mirror is a converging mirror , It is used as a torch to reflect the light ,

2-It is used in the aircraft landing at the airports to guide the aero ,
3-It is used in shaving to get an enlarged and erect image of the face

4-The concave mirror is used in front lights of cars to reflect the light , It is used in marine lighthouses that are found at the marine ports and at the airports to guide the ships and it is used in the solar ovens .
5-The concave mirror is used in the solar ovens and the solar furnaces to collect a large amount of solar energy in the focus of the mirror for cooking food , heating water , recharging power backups or melting metals, respectively .

6-Concave mirrors are used in satellite dishes , They are used in telescopes , Dentist and ENT doctors use them to obtain a larger image than the original of the teeth , ear or skin etc .
7-Concave mirrors are used in the electron microscopes and magnifying glasses , They are used in the visual bomb detectors, and they are used in the flashlight mirror of camera .

Why it is impossible to get a real image with a concave mirror?.

2-4-6-Application on concave mirror. (Fig. 2-9-a,b,c and d)

a)

b)

c)

d)

(Fig. 2-9-a, b, c and d)

2-4-6-Application on concave mirror. (Fig. 2-9-a,b,c and d)

a)

b)

c)

d)

(Fig. 2-9-a, b, c and d)

2-4-7-Special application on concave mirror:(Fig. 2-10-a,b,c,d and e.)

a)

b)

c)

d)

e)

(Fig. 2-10-a,b,c,d and e.)

2-4-8-Special application on concave mirror. (Fig. 2-11-a,b and c.)

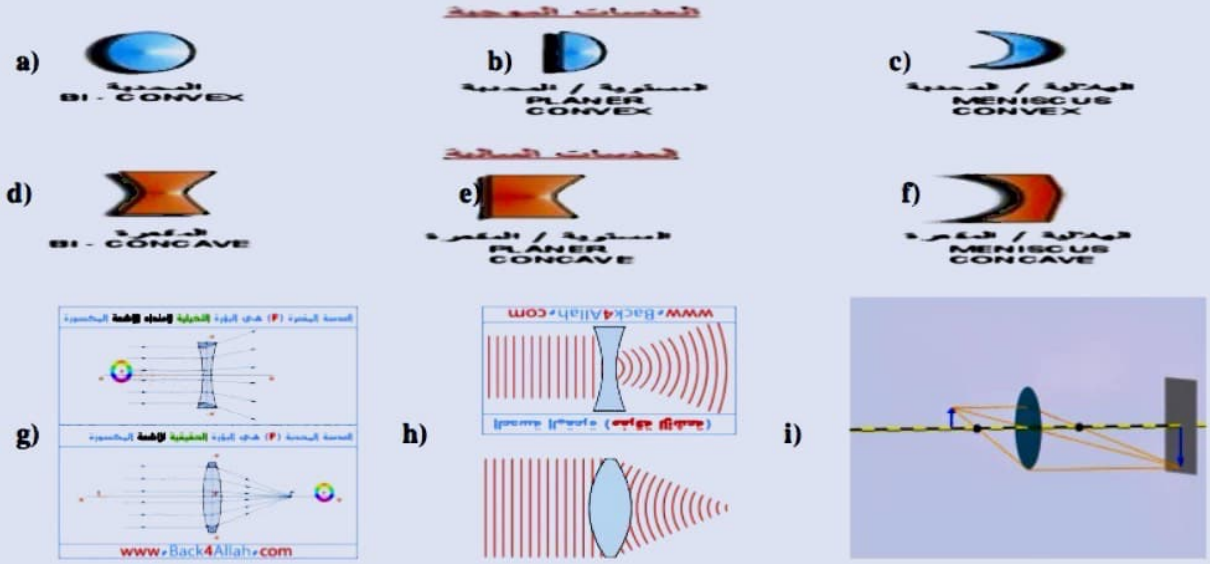
a)

b)

c)

(Fig. 2-11-a,b and c.)

2-5 -Lenses : General information about the lenses. (Fig. 2-12-a, b ,c, d, e, f, g, h and i .)



(Fig. 2-12-a, b ,c, d, e, f, g, h and i .)

Put True or False in front of the fo [redacted]

1- The mirrors laws are used by the scientists to invent a numerous application , such as the pictures from the Satellites , Radar and TV,..... .

2-Plane and spherical mirrors are used to form three dimensional images of two-dimensional objects.

3-If the size, orientation and location of an object relative to a mirror or a lens are known, the law of reflection and ray tracing can be used to locate the image graphically.

4-The convex mirror Is used by the dentist to see the lower parts of the dental.

5-When the object is put in ∞ (back of the mirror)the image will be in the focus.

Choose the correct answer from the following: [redacted]

1- Which one of the following pictures is related to the concave mirror ?

a)

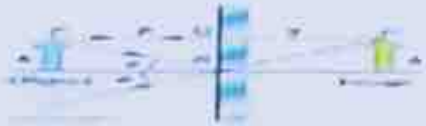
b)

c)

2-Which one of the following pictures is related to the convex mirror ?

a)

b)

c) 

3- The convex mirror are put on the corners of roads so that you can see any cars coming to avoid the collisions.

a)

b)

c)